

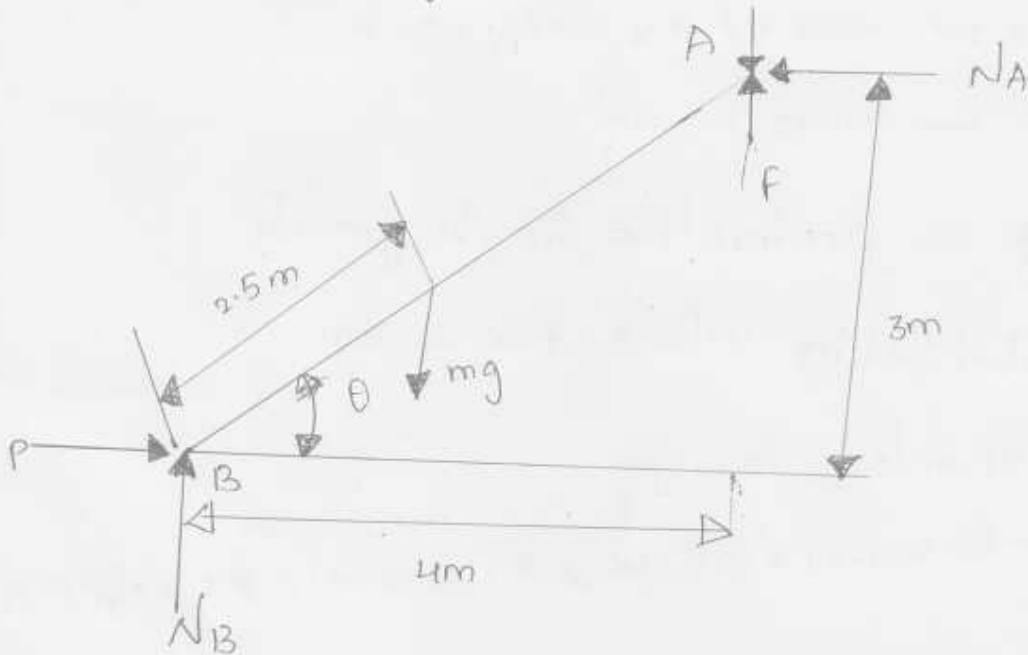
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Answer

Answer:

The free body diagram was



Contact surface at B is smooth

The angle of the rod with respect to horizontal

$$\tan \theta = \frac{3}{4}$$

$$\theta = \frac{3}{4} \tan^{-1}$$

$$\theta = 36.87^\circ$$

friction at A.

$$F = \mu_s N_A \text{ formula.}$$

μ_s is static friction

we should substitute 0.2

$$F = 0.2 N_A$$

moments at point B

$$\sum M_B = 0$$

$$-mg \times 2.5 \cos \theta + F \times 4 + N_A \times 3 = 0.$$

M — Mass of rod

g — Acceleration due to gravity

Substituting values. for 30 kg.

9.81 m/s^2 for g .

$$-30 \times 9.81 \times 2.5 \cos 36.87^\circ + 0.2 N_A \times 4 + N_A \times 3 = 0$$

$$-588.59 + 3.8 N_A = 0$$

$$3.8 N_A = 588.59$$

$$N_A = \frac{588.59}{3.8}$$

$$N_A = 154.89 \text{ N}$$

Equilibrium in x-direction.

$$\sum F_x = 0$$

$$P - N_A = 0 \quad [\because N_A = 154.89 \text{ N}]$$

Substituting values for N_A .

$$P - 154.89 \text{ N} = 0$$

$$P = 154.89 \text{ N}$$

$$\therefore P = 155 \text{ N}$$

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